

LEMLAND NYHAMN, Finland

WMO: 029800

Lat: 59.9593N

Lon: 19.9551E

Elev: 33

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.7	17.1	7.1	7.9	13.7	11.4	9.9	18.6	43.6	33.8	39.9	32.2	15.8	50	0.844

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	72.7	67.2	70.4	65.5	68.3	64.0	68.5	71.9	66.4	69.6	64.7	67.6	11.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	99.8	70.7	64.9	92.4	68.8	63.2	87.0	67.0	32.8	72.4	31.1	69.6	29.7	67.9	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.1	34.8	31.5	DB	11.5	75.4	8.2	4.0	5.6	78.3	0.8	80.7	-3.7	82.9	-9.7	85.9	(4)
(5)			WB	10.1	68.2	7.8	3.4	4.5	70.6	-0.1	72.6	-4.5	74.5	-10.2	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	31.4	29.5	32.2	38.1	46.0	54.5	62.5	62.8	55.9	47.1	40.2	35.6	(6)
(7)		DBStd	12.50	5.84	7.11	4.76	3.54	4.78	4.18	4.53	4.05	4.06	4.78	5.01	5.72	(7)
(8)		HDD50	3069	577	574	551	357	143	8	0	0	4	113	294	448	(8)
(9)		HDD65	7447	1042	994	1016	807	588	316	106	93	272	556	744	913	(9)
(10)		CDD50	1152	0	0	0	0	20	142	389	396	183	22	0	0	(10)
(11)		CDD65	53	0	0	0	0	0	0	29	24	0	0	0	0	(11)
(12)		CDH74	35	0	0	0	0	0	0	29	6	0	0	0	0	(12)
(13)		CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	16.6	19.4	17.7	16.6	15.0	14.0	14.0	13.5	15.0	16.4	18.2	19.5	20.5	(14)
(15)	Precipitation	PrecAvg	19.5	1.5	1.1	0.9	1.1	1.3	1.9	1.8	2.2	1.9	1.9	2.0	1.8	(15)
(16)		PrecMax	28.4	3.1	2.1	2.0	2.6	3.0	3.7	4.6	5.6	4.6	4.3	4.2	3.7	(16)
(17)		PrecMin	14.4	0.1	0.2	0.0	0.2	0.3	0.6	0.2	0.0	0.2	0.0	0.6	0.4	(17)
(18)		PrecStd	3.0	0.7	0.5	0.5	0.7	0.7	0.8	1.1	1.3	1.1	0.9	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	40.5	42.3	51.0	63.6	68.0	77.4	74.7	66.4	58.0	50.2	45.9	(19)
(20)			MCWB	40.6	38.0	38.6	44.6	55.0	59.1	69.4	68.6	63.2	55.8	48.6	43.9	(20)
(21)		2%	DB	40.6	39.1	40.1	47.0	58.7	64.4	73.7	71.9	64.6	56.1	48.9	44.5	(21)
(22)			MCWB	39.4	37.3	37.6	42.6	53.1	58.1	67.8	67.0	61.3	54.1	47.5	43.1	(22)
(23)		5%	DB	39.4	37.9	38.8	44.6	55.1	62.1	71.6	70.1	62.9	54.4	47.6	43.5	(23)
(24)			MCWB	38.0	36.5	36.7	40.9	50.6	57.3	66.6	65.0	59.8	52.4	46.2	42.0	(24)
(25)		10%	DB	38.3	36.9	37.6	42.7	52.3	60.2	69.3	68.3	61.4	53.1	46.4	42.2	(25)
(26)			MCWB	36.9	35.7	35.8	40.0	48.4	56.3	65.0	64.0	58.6	51.2	45.0	40.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	38.7	39.6	45.8	56.7	61.9	71.6	71.5	64.2	56.7	49.2	44.8	(27)
(28)			MCDWB	41.9	39.4	41.1	48.9	60.4	65.6	74.6	73.1	65.7	57.5	49.6	45.3	(28)
(29)		2%	WB	39.8	37.7	38.2	43.2	54.0	59.7	69.2	68.3	62.0	54.5	47.7	43.7	(29)
(30)			MCDWB	40.4	38.5	39.5	45.8	57.7	62.6	72.8	71.2	64.0	55.6	48.6	44.4	(30)
(31)		5%	WB	38.4	36.8	37.1	41.4	51.1	58.2	67.3	66.2	60.6	53.1	46.6	42.4	(31)
(32)			MCDWB	39.3	37.7	38.3	43.6	54.7	60.9	70.9	69.1	62.4	54.2	47.6	43.4	(32)
(33)		10%	WB	37.3	35.9	36.0	40.2	48.9	56.9	65.6	64.7	59.2	51.8	45.3	41.1	(33)
(34)			MCDWB	38.3	36.9	37.2	42.1	51.9	59.5	68.9	67.6	61.0	53.1	46.3	42.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	4.8	4.5	5.4	6.1	5.9	5.4	4.6	4.1	3.7	3.9	4.4	(35)
(36)			MCDBR	4.3	4.2	5.1	8.4	10.7	8.9	7.9	5.8	4.6	3.7	4.0	4.0	(36)
(37)			MCWBR	4.6	3.6	3.8	5.2	6.2	5.0	4.9	3.9	4.1	3.8	4.3	4.2	(37)
(38)		5% WB	MCDWB	4.3	3.8	4.7	7.2	9.7	7.1	6.7	5.1	4.1	3.5	3.7	3.7	(38)
(39)			MCWBR	4.6	3.6	3.8	5.0	6.4	4.8	4.9	4.2	4.1	3.8	4.1	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.313	0.333	0.371	0.368	0.359	0.388	0.378	0.348	0.334	0.304	N/A	(40)	
(41)		taud	2.186	2.259	2.321	2.338	2.423	2.489	2.430	2.445	2.514	2.418	2.231	N/A	(41)	
(42)		Ebn at Noon	157	216	252	263	273	277	265	259	249	210	143	N/A	(42)	
(43)		Edh at Noon	13	22	28	33	32	31	32	30	23	19	13	N/A	(43)	
(44)	All-Sky Solar Radiation	RadAvg	79	256	743	1281	1744	1932	1801	1407	874	373	107	51	(44)	
(45)		RadStd	10	47	81	121	146	121	162	119	95	57	24	7	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-193	N/A	N/A	N/A	(47)

Nomenclature: See separate page